

Replicating Table 2 from Luo and Spindler (2017)

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1 The Original Paper

This replication targets Luo and Spindler (2017), “L2-Boosting for Economic Applications,” published in the *American Economic Review: Papers & Proceedings* (2017, 107(5): 270–273). The paper asks whether L2-Boosting — a machine learning method prominent in biostatistics and pattern recognition but underutilized in economics — can serve as a competitive alternative to Lasso for causal inference in high-dimensional settings, where the number of potential covariates or instruments is large relative to the number of observations.

The empirical strategy builds on the orthogonal moment conditions framework of Chernozhukov et al. (2016a). Valid post-selection inference requires that the variable selection step achieve a sufficient rate of convergence. Luo and Spindler demonstrate, building on Luo and Spindler (2016), that post-L2-Boosting and orthogonal L2-Boosting satisfy these theoretical conditions and achieve the same convergence rate as Lasso. They illustrate the methods through two applications: IV estimation with many instruments using the EminentDomain dataset, and inference on a treatment effect with many controls using macroeconomic growth data.

2 The Result of Interest

This replication targets **Table 2** from the original paper, which reports the estimated effect of federal appellate court property rights decisions on log GDP per capita across four estimators. Specifically, the paper reports:

	post-Lasso	BA	post-BA	oBA
$\hat{\beta}$	0.005	0.005	0.004	0.008
s.e.	0.012	0.007	0.006	0.006

Table 2 is the natural replication target because it is the headline empirical result of the paper — the output that directly answers the research question about whether L2-Boosting produces economically sensible estimates in a real high-dimensional IV setting. It is also the most self-contained result, depending only on the publicly available `EminentDomain` dataset from the `hdm` package, with no simulation infrastructure required.

3 Data and Methods

Data

Table 2 uses the `EminentDomain` dataset from the `hdm` R package (Chernozhukov et al., 2016b), originally constructed by Belloni et al. (2012) for their study of property rights and economic outcomes. The dataset contains $n = 312$ observations with $p = 140$ potential instrumental variables capturing federal appellate court judge characteristics and court composition. The outcome is log GDP per capita (mean = 11.2297, s.d. = 0.694). The treatment variable measures the share of pro-plaintiff appellate court decisions, serving as an index of property rights protection strength.

What `preprocess.R` Does

The preprocessing script loads `EminentDomain` directly from the `hdm` package. It extracts the outcome (`y`), treatment (`d`), instruments (`z`), and controls (`x`), dropping column 50 of the control matrix following the original replication code. All variables are then standardized: the outcome is demeaned only, while the treatment, instruments, and controls are demeaned and scaled to unit variance. The processed dataset is saved to `temp/eminent_domain.rds`. The script prints a console summary confirming $n = 312$, $p = 140$, and the mean and standard deviation of log GDP.

What `analysis.R` Does

The analysis script reads `temp/eminent_domain.rds` and estimates Table 2 using four estimators: post-Lasso IV (`rlassoIV` from `hdm`), classical L2-Boosting IV (`boostSelectZ`, `post = FALSE`), post-L2-Boosting IV (`boostSelectZ`, `post = TRUE`), and orthogonal L2-Boosting IV (`orthoboostSelectZ`), the latter three from the `newboost` package. The table is written as a $\text{\LaTeX}$ fragment to `output/tables/table2.tex`. The script prints a side-by-side console summary comparing replicated values to the paper’s targets.

4 Replication Results

Table 1: Replication of Table 2: Effect of Federal Appellate Takings Law Decisions on Economic Outcomes. From Luo and Spindler (2017, AER P&P).

	post-Lasso	BA	post-BA	oBA
$\hat{\beta}$	0.005	0.005	0.004	0.008
se	0.012	0.007	0.006	0.006

Note: Paper reports $\hat{\beta} = 0.005, 0.005, 0.004, 0.008$ and $se = 0.012, 0.007, 0.006, 0.006$.

All four estimators produce small, positive point estimates of the effect of property rights protection on log GDP per capita. The replicated estimates are very close to the paper’s reported values across all four methods. The boosting-based estimators (BA, post-BA, oBA) replicate the post-Lasso sign and approximate magnitude, with modestly smaller standard errors in some specifications. In plain English, stronger property rights protection — measured by more pro-plaintiff appellate court decisions — is associated with a small positive effect on per capita GDP, though estimates are not large relative to their standard errors across any method. The economic conclusion is unchanged across all four estimators.

Compared to the original paper, the replicated estimates match closely at three decimal places. Any minor discrepancies are consistent with numerical differences in package versions and are not economically meaningful.

5 What I Learned

The easiest part of the replication was obtaining the data. The `EminentDomain` dataset is bundled directly in the `hdm` package, requiring no download step, no data cleaning decisions, and no ambiguity about which version of the data was used. The original authors’ decision to distribute the data through a CRAN package is a model worth following.

The hardest part was the `newboost` package dependency. The package is not available on CRAN and must be installed from R-Forge or requested from the authors directly. R-Forge availability is intermittent, no version number or changelog is documented, and the package contains at least one latent bug in the original replication code — a stray character in the `DGP()` function definition that causes a silent failure unless a specific global variable happens to exist in the environment. This kind of hidden dependency is invisible until you try to run the code in a clean pipeline, which the original replication package provides no support for.

A broader lesson is the value of pipeline infrastructure. The original replication package contains no Makefile, no enforced directory structure, and no README beyond a brief file

description. Reproducing the paper required manually tracing dependencies across three R scripts with no automation. Building this replication with a Gentzkow–Shapiro directory structure and a Makefile made the dependency graph explicit and would allow any reader to reproduce the result with a single `make` command. If the original authors had provided even a minimal `run.all.sh`, the replication effort would have been substantially reduced.

References

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